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Odoo V18 Technical Training

Prasetiya Mulya University

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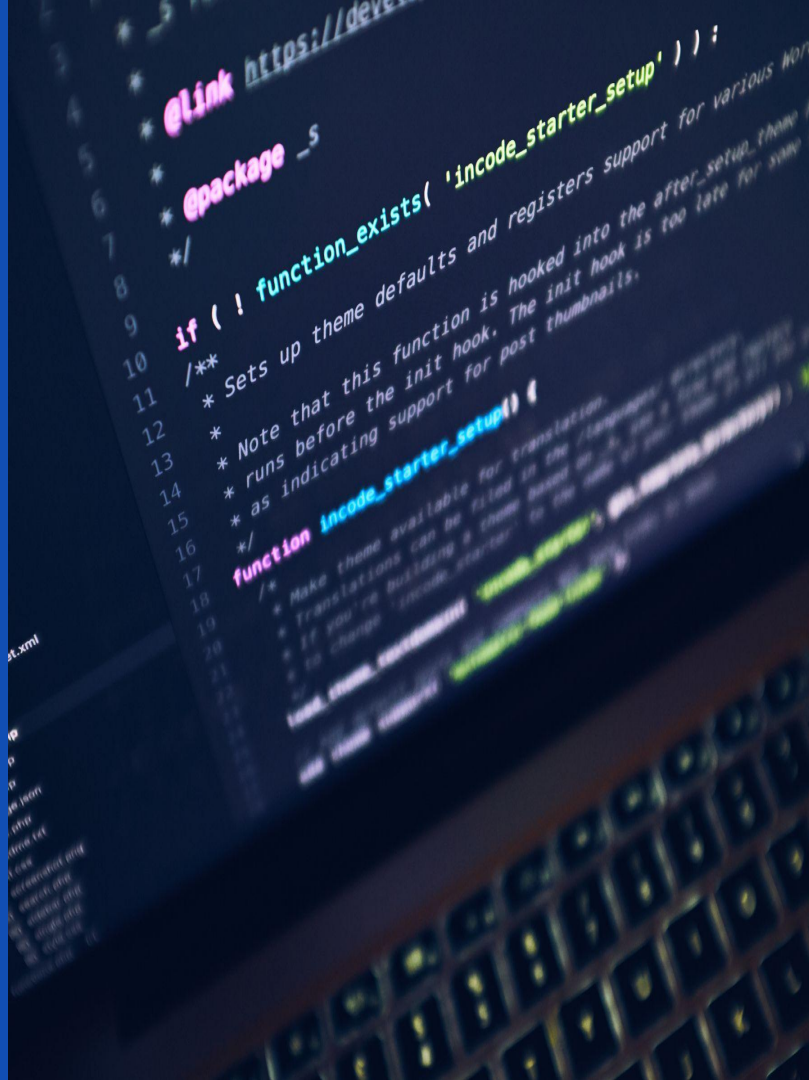
ODOO 18 · TECHNICAL FUNDAMENTAL

Day 5 — Odoo as REST API Provider

HTTP Controller · Serializer · Response Contract · API Key

Academy Management · studi kasus berkelanjutan dari Day 4 final



Odoo Trainer Technical

Ardian Pramana
Lead Developer @Arkana

Rundown **Training**
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08.30 - 10.00	: Sesi I
10.00 – 10.15	: Coffee Break
10.15 – 12.00	: Sesi II
11.30 – 13.15	: ISHOMA
13.15 – 15.20	: Sesi III
15.20 – 15.45	: Break Ashr
15.45 – 16.30	: Sesi IV

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DAY 5 · OVERVIEW

What We Are Building

GET	/academy/api/v1/ping	Health check	A
GET	/academy/api/v1/courses	Daftar course (serialized)	B
GET	/academy/api/v1/courses/<code>	Course detail + 404	C
POST	/academy/api/v1/enrollment-requests	Enrollment draft + API Key	D+E

⚠️ **Enrollment selalu DRAFT — approval workflow tetap berjalan. Tidak ada auto-create batch. API tidak bypass ORM.**

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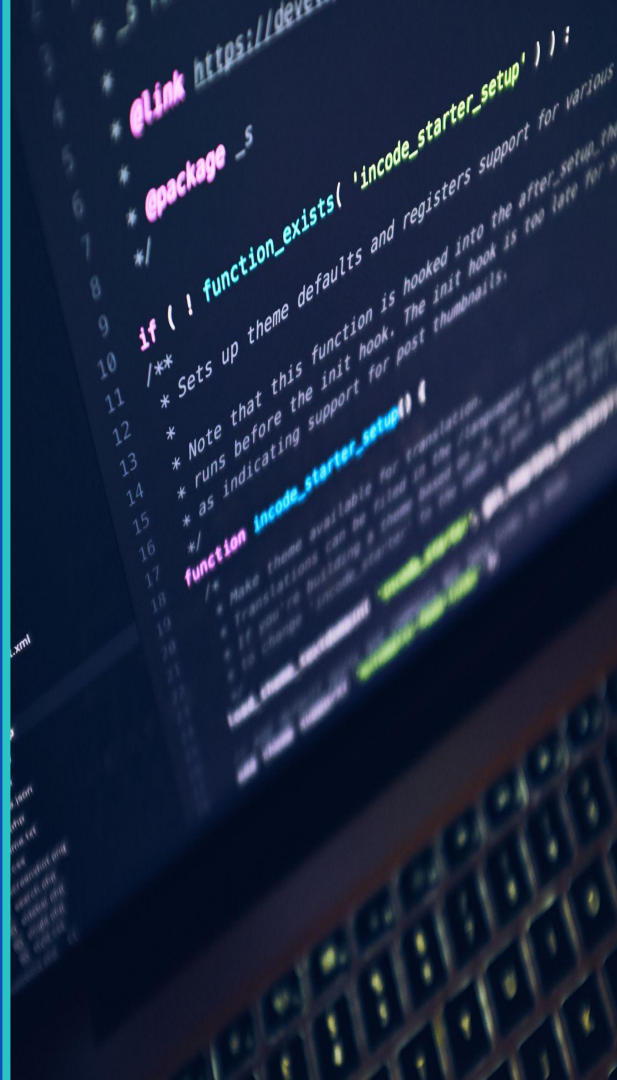
PART 1

How REST Provider Works in Odoo

Odoo model tidak otomatis jadi REST resource



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PART 1 · CONCEPT

How REST Provider Works in Odoo

- Controller = API boundary, bukan tempat business logic.
- Business rules tetap di Model / Workflow. Serializer mengontrol field yang aman.
- API tidak pernah bypass approval workflow — enrollment selalu dibuat draft.

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PART 1 · CONCEPT

Controller sebagai API Boundary

✅ Controller BOLEH

- Parse request payload / path params
- Validasi field wajib ada atau tidak
- Lookup record via ORM (env[...])
- Panggil business method di Model
- Serialize response ke JSON dictionary
- Return HTTP status code yang tepat

❌ Controller JANGAN

- Menulis business rule langsung di controller
- Bypass approval workflow
- Auto-confirm enrollment tanpa review
- Return seluruh recordset fields (tanpa filter)
- Hardcode credentials di kode
- Return stack trace ke client

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PART 1 · CONCEPT

Anatomy of Odoo HTTP Controller

Extend http.Controller
Bukan class biasa

@http.route()
Mendaftarkan URL ke Odoo router

type="http"
Return Response manual, bukan JSON-RPC

auth="public"
Tidak butuh Odoo session (+ sudo di dalam)

request.make_response()
Kontrol penuh atas status & body

```
from odoo import http
from odoo.http import request
import json

class AcademyApiController(http.Controller):

    @http.route(
        "/academy/api/v1/ping",
        type="http",       # ← bukan "json"
        auth="public",
        methods=["GET"],
    )
    def ping(self, **kw):
        payload = {
            "success": True,
            "data": {"status": "ok", "version": "1.0"},
            "error": None,
        }
        return request.make_response(
            json.dumps(payload),

headers=[("Content-Type","application/json")],
            status=200,
        )
```

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PART 1 · CONCEPT

Mengapa type="http" untuk Lab Ini?

type="json" (JSON-RPC)

- Request harus format {jsonrpc, method, params}
- Response dibungkus {result: ..., id: ...}
- HTTP status selalu 200
- Tidak cocok untuk REST-style API

type="http" ← kita pakai ini

- Request: plain HTTP (GET, POST, dll.)
- Response: bebas, kita kontrol penuh
- HTTP status kita tentukan (200, 201, 400, 404…)
- Cocok untuk REST API provider

type="http" + make_response(json.dumps(...), status=...) = REST API pattern kita hari ini.

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PART 1 · CONCEPT

Response Contract

Semua endpoint menggunakan shape yang sama.

HTTP 200 / 201

```
// Success
{
  "success": true,
  "data": { ... },
  "error": null
}
```

HTTP 4xx

```
// Error
{
  "success": false,
  "data": null,
  "error": {
    "code": "ERROR_CODE",
    "message": "Readable message."
  }
```

HTTP Status	Kapan dipakai
200 OK	GET berhasil; POST → enrollment sudah ada
201 Created	POST → enrollment baru berhasil dibuat
400 Bad Request	Field wajib tidak ada / JSON tidak valid
401 Unauthorized	API Key salah atau tidak ada

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PART 1 · CONCEPT

Serializer & Safe Field Exposure

```
def _course_to_dict(self, course):
    # Only expose safe primitive fields.
    # description = fields.Html() — excluded.
    return {
        "code":           course.code or "",
        "name":           course.name,
        "level":          course.level or "",
        "duration_hours": course.duration_hours,
    }
```

- ✅ Whitelist fields secara eksplisit
- ✅ Method terpisah, bukan inline dict
- ✅ Return primitive (str, int, bool, float)
- ❌ Jangan return recordset langsung
- ❌ Jangan expose Html fields ke JSON
- ❌ Jangan invent fields yang tidak ada

Serializer adalah "export schema". Definisikan eksplisit — jangan andalkan read([]) tanpa filter.

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LAB CHECKPOINT A

Checkpoint A — Ping Endpoint

GET /academy/api/v1/ping

1. Buat folder controllers/ di dalam module academy_management
2. Buat controllers/__init__.py: from . import api_controller
3. Edit root __init__.py: tambah from . import controllers
4. Buat class AcademyApiController(http.Controller)
5. @http.route('/academy/api/v1/ping', type='http', auth='public', methods=['GET'])
6. Return make_response(json.dumps({"success":True,"data":{...},"error":None}), status=200)

```
$ curl -s http://localhost:8069/academy/api/v1/ping
→  {"success": true, "data": {"status": "ok", "version": "1.0"}, "error": null}
```

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LAB CHECKPOINT B

Checkpoint B — GET Courses List

```
@http.route("/academy/api/v1/courses",
            type="http", auth="public",
            methods=["GET"])
def get_courses(self, **kw):
    courses = request.env["academy.course"] \\
                    .sudo().search([])
    return _ok([self._course_to_dict(c)
                for c in courses])

def _course_to_dict(self, course):
    return {
        "code":           course.code or "",
        "name":           course.name,
        "level":          course.level or "",
        "duration_hours": course.duration_hours,
    }
```

sudo()
auth='public' → pakai sudo untuk ORM access

_course_to_dict()
Serializer — tidak return recordset langsung

list comprehension
Iterasi semua course, serialize satu per satu

_ok() helper
Return {"success":true,"data":[...],"error":null}

description = Html
Tidak dimasukkan serializer — tidak JSON-safe

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LAB CHECKPOINT C

Checkpoint C — GET Course Detail + 404

```
@http.route("/academy/api/v1/courses/<string:code>",
            type="http", auth="public",
            methods=["GET"])
def get_course(self, code, **kw):
    course = request.env["academy.course"] \\
                  .sudo().search(
                      [("code", "=", code)], limit=1)

    if not course:
        return _err(
            "COURSE_NOT_FOUND",
            "Course with code '%s' not found." % code,
            status=404,
        )

    return _ok(self._course_to_dict(course))
```

<string:code>
Dynamic route param — Odoo router extract otomatis

search([...], limit=1)
Cari berdasarkan code, limit 1

if not course:
Recordset kosong = falsy — cek ini

_err() helper
Return {"success":false,"data":null,"error":{...}}

status=404
HTTP 404 = transport; COURSE_NOT_FOUND = aplikasi

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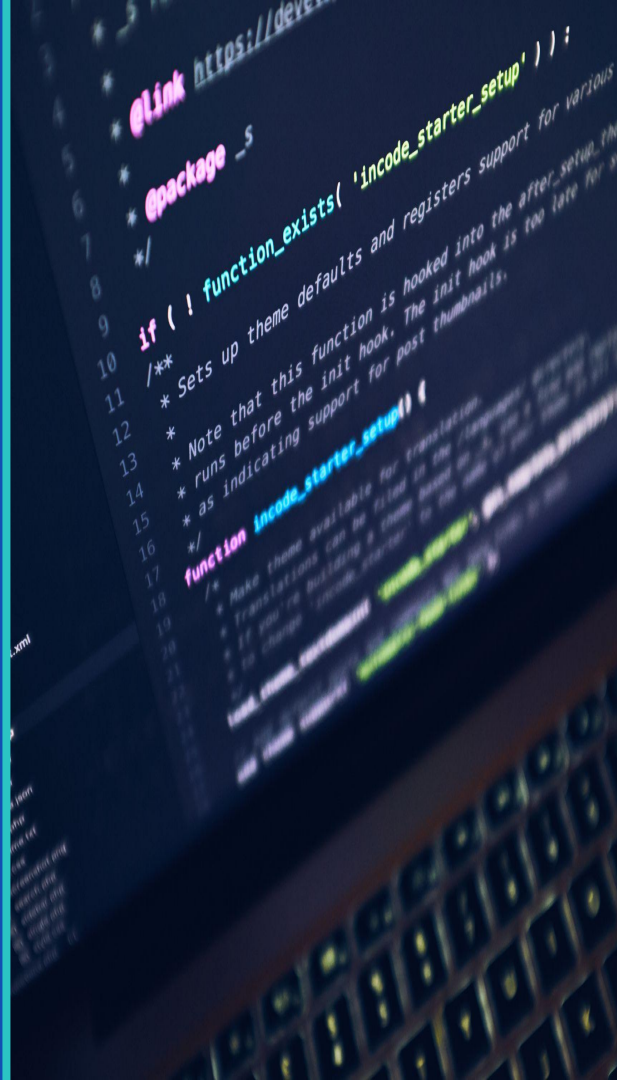
PART 2

Business-safe POST Endpoint

Enrollment selalu draft — approval workflow tidak disentuh



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PART 2 · CONCEPT

Business-safe POST Endpoint

```
POST
/academy/api/v1/enrollment-requests
{
  "student_name":  "Budi Santoso",
  "student_email": "budi@example.com",
  "batch_code":    "PY-101-JAN",
  "notes":         "Weekend batch"
}
```

1. Parse & validate payload
2. Lookup batch by batch_code
3. Find/create student by email
4. Create enrollment → state='draft'
5. **Existing → 200 OK | New → 201 Created**

⚠️ **Tidak ada auto-create batch. Tidak ada state=confirmed. Tidak ada set approval metadata.**

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PART 2 · CONCEPT

Mengapa batch_code? Tidak Auto-create Batch?

batch_code sebagai External Key

- External system tidak tahu internal Odoo ID
- code adalah identifier stabil yang bisa dishare
- URL-friendly dan human-readable (PY-101-JAN)
- Unique constraint wajib ada di academy.batch.code

```
_sql_constraints = [
    ("code_unique", "unique(code)",
     "Batch code must be unique."),
]
```

Tidak Auto-create Batch

- Batch dibuat melalui proses bisnis (approval, kapasitas)
- API hanya boleh reference batch yang sudah ada
- Jika tidak ditemukan → 404 BATCH_NOT_FOUND
- API tidak punya konteks untuk create batch baru

💡 **API yang baik: lookup yang sudah ada, jangan ciptakan data tanpa konteks bisnis. Resource tidak ditemukan → 404.**

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LAB CHECKPOINT D

Checkpoint D — POST Enrollment Request

```
@http.route("/academy/api/v1/enrollment-requests",
    type="http", auth="public",
    methods=["POST"], csrf=False)
def create_enrollment_request(self, **kw):
    try:
        payload = json.loads(
            request.httprequest.data or b"{}")
    except (json.JSONDecodeError, ValueError):
        return _err("INVALID_JSON","...",status=400)
    required = ["student_name","student_email","batch_code"]
    missing = [f for f in required
               if not (payload.get(f) or "").strip()]
    if missing:
        return _err("MISSING_FIELDS","...",status=400)
    batch = env["academy.batch"].sudo().search(
        [("code","=",payload["batch_code"])], limit=1)
    if not batch:
        return _err("BATCH_NOT_FOUND","...",status=404)
    ...create student if needed...
    if existing: return _ok({...}, status=200)
    enrollment = env["academy.enrollment"].sudo().create({
        "batch_id": batch.id, "student_id": student.id,
        "notes": notes,  # bukan "note"
```

csrf=False
Wajib untuk API — tidak ada browser form

json.loads(...data)
Read raw body, bukan kw params

MISSING_FIELDS
Validate sebelum ORM access

BATCH_NOT_FOUND
Lookup dulu — jangan auto-create

"notes" bukan "note"
academy.enrollment.notes = fields.Text()

201 vs 200
Baru=201 Created; Existing=200 OK

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LAB CHECKPOINT E

Checkpoint E — API Key Boundary

POST request → Header X-API-Key → Validated → Proceed (or 401)

```
def _check_api_key(self):
    # Key name  : academy_management.api_key
    # Demo value: academy-demo-key
    #             (seeded in data/academy_api_data.xml)
    incoming = request.httprequest.headers.get(
        "X-API-Key", "")
    stored = (
        request.env["ir.config_parameter"]
        .sudo()
        .get_param("academy_management.api_key", "")
    )
    return bool(incoming and incoming == stored)

# In create_enrollment_request() — first line:
if not self._check_api_key():
    return _err("UNAUTHORIZED",
                status=401)
```

⚠️ **Shared Key**
Tidak terikat user Odoo spesifik

⚠️ **No Audit Log**
Tidak ada trail siapa yang memanggil

⚠️ **Plain Comparison**
Tidak ada hashing / timing-safe

✅ **Cukup untuk Lab**
Pola boundary — bukan prod auth

⚠️ **Demo simplification. Production: res.users.apikeys atau OAuth2, bukan shared key.**

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PART 2 · SECURITY

⚠️ auth='public' + sudo() — Pahami Risikonya

Aspek	Artinya	Risiko
auth='public'	Request tidak butuh session Odoo	Siapapun bisa akses URL — belum ada gate
sudo()	ORM berjalan dengan hak superuser	Bypass Odoo access rights — berbahaya tanpa validasi
Kombinasi keduanya	Tidak perlu login, akses semua data	Production: WAJIB punya gate (API Key, OAuth)
Lab context	Training simplification	Di produksi: gunakan user-scoped API key / OAuth2

Production Direction:

→ Gunakan res.users.apikeys (Odoo native) atau OAuth2

→ Scope akses per user / partner — bukan shared superuser

→ Log semua request (siapa, kapan, apa yang diakses)

→ Rate limit di layer Nginx/reverse proxy

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LAB · TEST

Postman Test Plan

1	GET	GET Ping	200 success:true, data.status:ok
2	GET	GET Courses (all)	200 data:[...]
3	GET	GET Course (valid code)	200 course object
4	GET	GET Course (404)	404 COURSE_NOT_FOUND
5	POST	POST Enrollment (valid)	201 state:draft
6	POST	POST same payload again	200 'Enrollment already exists.'
7	POST	POST Missing field	400 MISSING_FIELDS
8	POST	POST Wrong API Key	401 UNAUTHORIZED
9	POST	POST No API Key header	401 UNAUTHORIZED
10	POST	POST Batch not found	404 BATCH_NOT_FOUND
11	POST	POST Invalid JSON	400 INVALID_JSON

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COMMON MISTAKES

Kesalahan yang Sering Terjadi

✗ Lupa "from . import controllers" di __init__.py	✓ Tambahkan baris tersebut — route tidak terdaftar tanpa ini
✗ type="json" di route	✓ Pakai type="http" agar bisa kontrol HTTP status code
✗ json.dumps(courses) langsung	✓ Recordset tidak serializable — selalu lewat _course_to_dict()
✗ Lupa csrf=False di POST route	✓ Wajib untuk API — tanpa ini POST selalu 403
✗ Pakai "note" di enrollment create	✓ academy.enrollment punya "notes" (fields.Text()) — bukan "note"
✗ Auto-create batch jika tidak ditemukan	✓ Return BATCH_NOT_FOUND 404 — jangan create batch
✗ Set state="confirmed" di POST	✓ Hanya default "draft" — approval workflow di Odoo UI
✗ Hardcode API key di source code	✓ Baca dari ir.config_parameter via get_param()

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DAY 5 · PRODUCTION

Production Awareness — Dari Lab ke Prod

Kode lab ini mendemo pola. Sebelum ke produksi:

Area	Lab (Sekarang)	Production Direction
Authentication	X-API-Key via ir.config_parameter	res.users.apikeys atau OAuth2
Access Control	auth='public' + sudo() global	Scope per user, with_user(user)
Error Handling	Basic try/except	Structured logging, no stack trace
Input Validation	Manual field check	Schema validator (Cerberus, dll.)
Rate Limiting	Tidak ada	Nginx limit_req
Audit Trail	Tidak ada	Log model / chatter / SIEM

⚠️ **Ini bukan scope hari ini — tapi Anda harus tahu ke mana arahnya.**

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DAY 5 · RECAP

Academy dengan Feature REST API

A Controller Boundary

http.Controller memisahkan HTTP dari ORM. Bukan tempat bisnis logic.

B Validasi Sebelum ORM

Cek field, cek resource — sebelum menyentuh database.

C Serializer Eksplisit

_course_to_dict() — hanya field aman yang keluar. Tidak return recordset.

D Response Contract

{success, data, error} — semua endpoint shape yang sama.

E Security Boundary (Demo)

X-API-Key via academy_management.api_key. Production butuh lebih.

F Approval Workflow Aman

Enrollment dibuat state='draft'. API tidak bypass approval Odoo.

Semua endpoint berjalan di atas approval workflow Day 3 final — API tidak pernah bypass. 🎉

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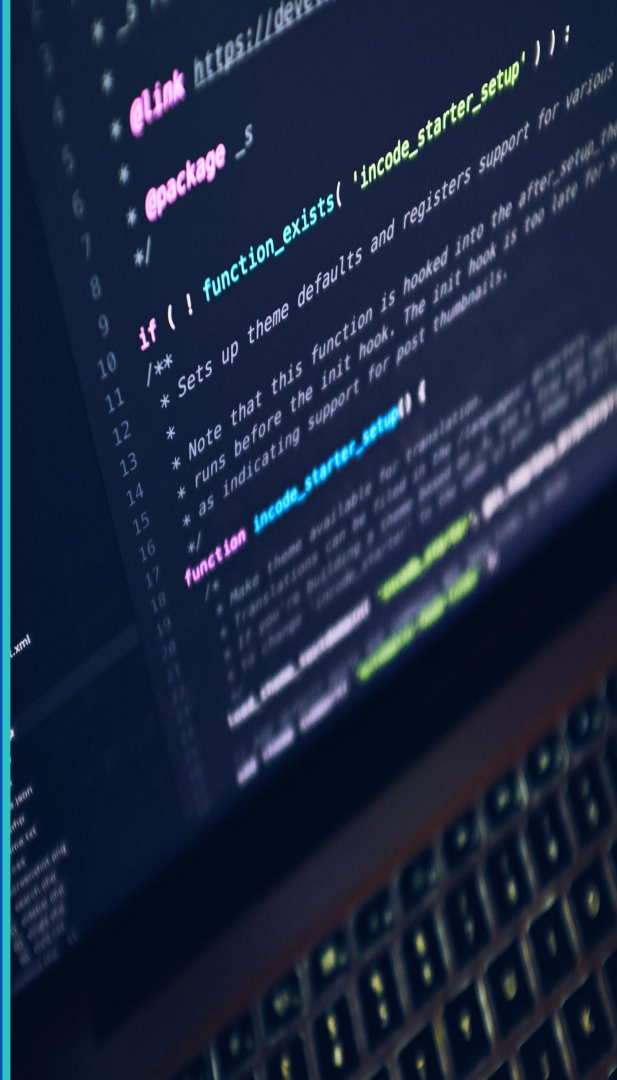
PART 3

Deployment

Panduan deployment untuk environment Prasmul



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DEPLOYMENT · FLOW

Satu fitur melewati tiga lingkungan

- Pilih entitas sebelum menentukan branch dan database.
- Feedback selalu kembali ke feature branch yang sama.
- Tidak ada direct push ke sandbox atau production.

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DEPLOYMENT · ENTITAS

Satu repository, dua jalur deployment

Tentukan entitas sebelum coding—jangan mencampur branch dan database.

Area	Universitas	ELI
Branch awal	production_univ	production
Local DB	Salinan DB Universitas	Salinan DB ELI
Sandbox	sandbox-univ	Belum dikonfirmasi
Production	production_univ	production
Server owner	Tim Prasmul	Tim Prasmul
Status	Jalur siap digunakan	Konfirmasi sandbox

⚠ **Branch 18.0-*-staging-support adalah target internal Arkana, bukan target PR tim Prasmul.**

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DEPLOYMENT · EXAMPLE

Contoh: fitur baru untuk Universitas

```
git switch production_univ
git pull
git switch -c feature-xyz

Local DB: v18_prasmul_univ_dev
Sandbox: sandbox-univ
Production: production_univ
```

1. Buat feature-xyz dari production_univ
2. Coding dan tes awal pada DB local Univ
3. PR feature-xyz → sandbox-univ
4. Perbaiki branch sampai testing disetujui
5. **PR feature-xyz → production_univ**

⚠️ **Reviewer Arkana yang melakukan merge—peserta tidak direct push.**

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DEPLOYMENT · GOVERNANCE

Triage: kontribusi tanpa akses merge

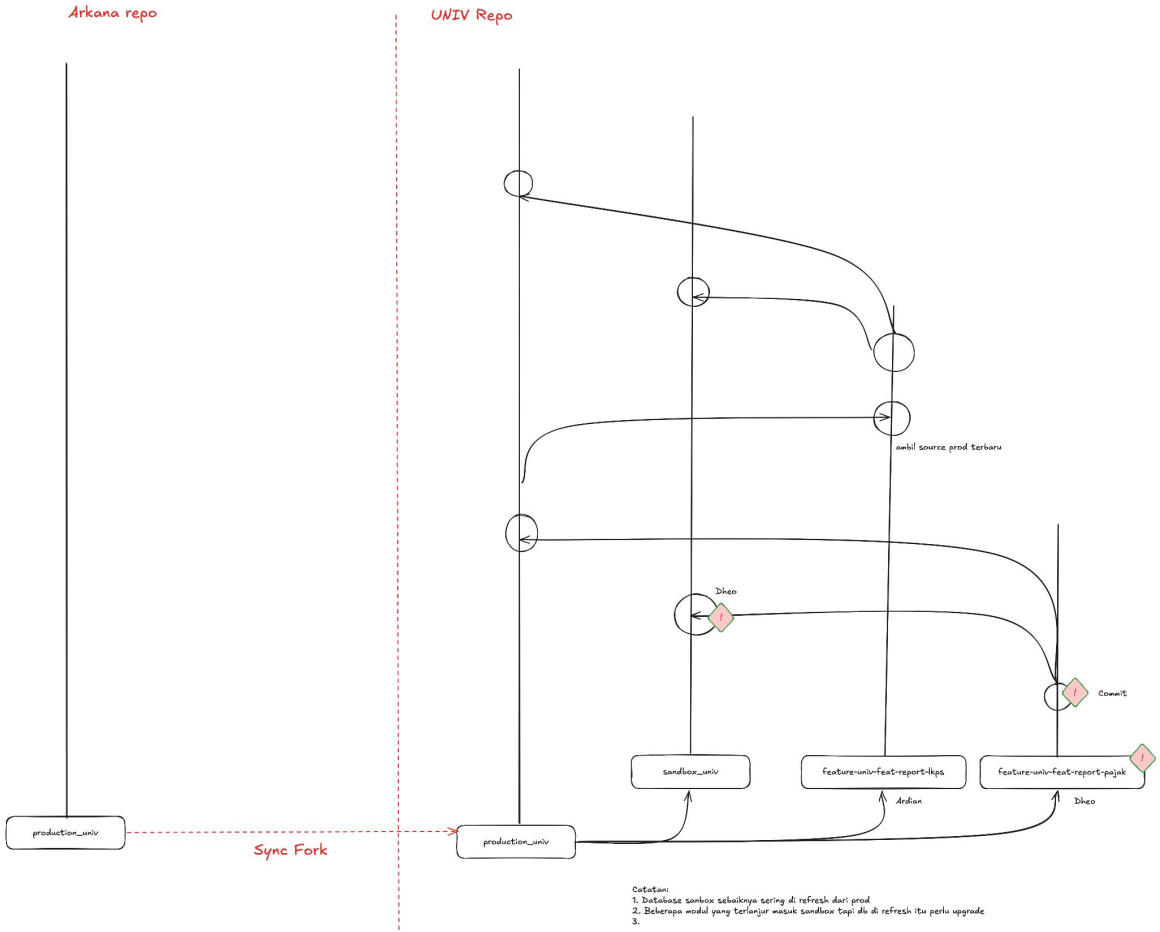
✅ PESERTA BOLEH

- Membuat fork dan feature branch
- Mengembangkan dan menguji di local
- Membuka PR ke sandbox
- Menindaklanjuti feedback di branch
- Melampirkan hasil testing
- Membuka PR production setelah disetujui

❌ PESERTA JANGAN

- Direct push ke branch utama
- Merge pull request sendiri
- Mencampur perubahan Univ dan ELI
- Mengganti target branch tanpa review
- Menguji dengan database entitas yang salah
- Deploy di luar proses yang disepakati

Corat-coret alur kerja git branching



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Q & A

Terima kasih telah menyelesaikan sesi development Training.

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Terima Kasih

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